

Unit 9

Symmetric Maps and Quadratic Forms

1. Quadratic forms

A real symmetric matrix A defines the quadratic form $q_A(x) = x^\top Ax = \sum_{i,j} a_{ij}x_i x_j$. The coefficient of $x_i x_j$, $i \neq j$, is $2a_{ij}$. For example,

$$3x^2 - 2xy + 2y^2 = \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Conversely $3x^2 + 5xy - 4y^2$ has symmetric matrix $\begin{pmatrix} 3 & 5/2 \\ 5/2 & -4 \end{pmatrix}$. For arbitrary real square B , $x^\top Bx = x^\top (B + B^\top)x/2$, because transposing the scalar $x^\top Bx$ gives $x^\top B^\top x$. Thus quadratic forms depend only on the symmetric part.

Theorem 9.1 (An extremum on the sphere is an eigenvector). Let $A = A^\top$. If u maximizes or minimizes q_A on $\{x : \|x\| = 1\}$, then $Au = \lambda u$, with $\lambda = u^\top Au$. In particular, a real symmetric matrix has a nonzero real eigenvector.

Proof. We use the analysis fact that a continuous function on the unit sphere attains a maximum and a minimum; the sphere is compact. For $n = 1$, every nonzero vector is already an eigenvector. For $n > 1$, take any unit $w \perp u$ and define $c(t) = u \cos t + w \sin t$. Then $\|c(t)\|^2 = \cos^2 t + \sin^2 t = 1$, $c(0) = u$, and $c'(0) = w$. Thus $g(t) = c(t)^\top A c(t)$ has an extremum at zero, and

$$0 = g'(0) = w^\top Au + u^\top Aw = 2w^\top Au,$$

where symmetry is essential for the last equality. Every vector in $u^\perp$ is zero or a scalar multiple of such a w . Hence $Au \perp u^\perp$, so $Au \in \text{span } u$. Write $Au = \lambda u$ and multiply by $u^\top$ to get $\lambda = u^\top Au$. $\square$

The vector w is tangent to the sphere and perpendicular to its radius u . This proof uses curves on the sphere, not an unproved multiplier rule.

Example 9.1 (A quadratic form on the unit circle). For $q(x, y) = 2x^2 - 3xy + y^2$,

$$A = \begin{pmatrix} 2 & -3/2 \\ -3/2 & 1 \end{pmatrix}, \quad p_A(t) = t^2 - 3t - \frac{1}{4}.$$

The eigenvalues are $\lambda_\pm = (3 \pm \sqrt{10})/2$. Set $v_\lambda = (1, \frac{2}{3}(2 - \lambda))^\top$; the first row of $Av = \lambda v$ holds by construction, and the second follows from $p_A(\lambda) = 0$. Both eigenvalues are simple, so all unit eigenvectors are

$$\pm \frac{v_{\lambda_+}}{\|v_{\lambda_+}\|}, \quad \pm \frac{v_{\lambda_-}}{\|v_{\lambda_-}\|}.$$

At a unit eigenvector, $q = \lambda$. The maximum is therefore λ_+ , attained at both signs of its unit eigenvector, and the minimum is λ_- , likewise attained at both signs.

2. Symmetric maps and invariant complements

On a finite-dimensional real inner-product space, a map T is *self-adjoint*, or symmetric, if $\langle Tv, w \rangle = \langle v, Tw \rangle$ for all v, w .

Proposition 9.2 (Adjoint in an inner-product space). Every linear map $T : V \rightarrow V$ has a unique adjoint T^* , satisfying

$$\langle Tv, w \rangle = \langle v, T^*w \rangle.$$

In an orthonormal basis, its matrix is the transpose of the matrix of T . Thus T is self-adjoint precisely when that matrix is symmetric.

Proof. Choose an orthonormal basis, so that the inner product of coordinate columns is the ordinary dot product. If A represents T , then $(Av)^\top w = v^\top A^\top w$; the map represented by $A^\top$ is therefore an adjoint. If S_1, S_2 both satisfy the identity, then $\langle v, (S_1 - S_2)w \rangle = 0$ for every v, w . Choosing $v = (S_1 - S_2)w$ proves $S_1 = S_2$. This uniqueness also makes the definition independent of the chosen orthonormal basis. $\square$

In a nonorthonormal basis, ordinary matrix transposition alone need not represent the adjoint: the inner product also has coordinates.

Proposition 9.3 (Invariant orthogonal complements). If $T = T^*$ and $TW \subseteq W$, then $T(W^\perp) \subseteq W^\perp$. Eigenvectors for distinct eigenvalues are orthogonal.

Proof. For $z \perp W$ and $w \in W$, $\langle Tz, w \rangle = \langle z, Tw \rangle = 0$, so $Tz \perp W$. If $Tv = \lambda v$ and $Tw = \mu w$, then

$$\lambda \langle v, w \rangle = \langle Tv, w \rangle = \langle v, Tw \rangle = \mu \langle v, w \rangle.$$

When $\lambda \neq \mu$, the inner product is zero. $\square$

Theorem 9.4 (Real spectral theorem). Every self-adjoint map on a finite-dimensional real inner-product space has an orthonormal eigenvector basis. Equivalently, for real $A = A^\top$, there exist real orthonormal columns $q_1, \dots, q_n$ and real numbers $\lambda_1, \dots, \lambda_n$ such that

$$Q^\top A Q = \Lambda, \quad A = Q \Lambda Q^\top = \sum_{j=1}^n \lambda_j q_j q_j^\top.$$

Proof. Induct on $\dim V$. The zero-dimensional case has an empty basis. For positive dimension choose an orthonormal coordinate basis. The sphere theorem supplies a real eigenvector, which we normalize to q_1 . The line span q_1 is invariant, so its orthogonal complement W , of dimension one less, is invariant. The restricted map $T|_W : W \rightarrow W$ is again self-adjoint, since the inner-product identity restricts to vectors in W . By induction W has an orthonormal eigenbasis $q_2, \dots, q_n$. Together with q_1 it gives the required basis. For matrices, $AQ = Q\Lambda$ and $Q^{-1} = Q^\top$ give the formulas. $\square$

Every algebraic multiplicity equals the dimension of its eigenspace, because diagonalization supplies exactly that many basis columns.

Example 9.2 (A complete three-dimensional spectral decomposition). Let

$$L = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}.$$

Expansion in row 1 of $tI - L$ gives

$$p_L(t) = (t-1)\{(t-2)(t-1)-1\} - (t-1) = t(t-1)(t-3).$$

Solving the three homogeneous systems gives

$$q_0 = \frac{1}{\sqrt{3}}(1, 1, 1)^\top, \quad q_1 = \frac{1}{\sqrt{2}}(1, 0, -1)^\top, \quad q_3 = \frac{1}{\sqrt{6}}(1, -2, 1)^\top.$$

Indeed $L(1, 1, 1)^\top = 0$, $L(1, 0, -1)^\top = (1, 0, -1)^\top$, and $L(1, -2, 1)^\top = 3(1, -2, 1)^\top$. Their pairwise dot products are zero, and their squared lengths before normalization are 3, 2, 6. Writing $Q = (q_0 \ q_1 \ q_3)$, we have $Q^\top LQ = \text{diag}(0, 1, 3)$. More explicitly,

$$L = \frac{1}{2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{pmatrix}.$$

This verifies the decomposition entry by entry.

3. Rayleigh quotients and equality cases

For real symmetric A and $x \neq 0$, define $R_A(x) = x^\top Ax / (x^\top x)$. It is unchanged when x is multiplied by a nonzero real scalar.

Theorem 9.5 (Rayleigh bounds). Order the eigenvalues as $\lambda_1 \geq \dots \geq \lambda_n$. Then

$$\lambda_n \leq R_A(x) \leq \lambda_1.$$

Equality on the right holds exactly for nonzero vectors in E_{λ_1} ; equality on the left holds exactly in E_{λ_n} . Thus the largest and smallest eigenvalues are the maximum and minimum of $x^\top Ax$ on the unit sphere.

Proof. Write $x = \sum_j c_j q_j$ in an orthonormal eigenbasis. Then $x^\top x = \sum_j c_j^2$ and $x^\top Ax = \sum_j \lambda_j c_j^2$. The quotient is a weighted average of the eigenvalues, with nonnegative weights $c_j^2 / \sum_i c_i^2$ summing to one. Moreover

$$\lambda_1 - R_A(x) = \frac{\sum_j (\lambda_1 - \lambda_j) c_j^2}{\sum_j c_j^2}.$$

It vanishes precisely when every coefficient belonging to a smaller eigenvalue is zero. This is exactly the top eigenspace. The lower bound and its equality condition follow in the same way. $\square$

Example 9.3 (The quotient and its restriction for L). For $x = (x_1, x_2, x_3)^\top$,

$$x^\top Lx = (x_1 - x_2)^2 + (x_2 - x_3)^2.$$

Hence the quotient is zero exactly for nonzero constant vectors. The spectral decomposition gives

$$0 \leq \frac{(x_1 - x_2)^2 + (x_2 - x_3)^2}{x_1^2 + x_2^2 + x_3^2} \leq 3,$$

with upper equality exactly on $\text{span}(1, -2, 1)^\top$. If $x_1 + x_2 + x_3 = 0$, the q_0 coefficient vanishes, so instead

$$1 \leq R_L(x) \leq 3.$$

The lower equality is attained on $\text{span}(1, 0, -1)^\top$. For the concrete $x = (1, -1, 0)^\top$, the quotient is $(4 + 1)/2 = 5/2$, between these bounds.

4. Positive forms and Gram matrices

A real symmetric matrix is *positive semidefinite* (PSD) if $x^\top Ax \geq 0$ for every x , and *positive definite* (PD) if $x^\top Ax > 0$ for every nonzero x . Write $A \succeq 0$ and $A \succ 0$, respectively; write $A \succeq B$ when $A - B \succeq 0$. The order notation is only used for symmetric matrices. A form is negative (semi)definite when its negative is positive (semi)definite; it is indefinite when it takes both positive and negative values.

Proposition 9.6 (Spectral tests and square roots). For real symmetric A , positivity of all eigenvalues is equivalent to positive definiteness; nonnegativity of all eigenvalues is equivalent to positive semidefiniteness. If $A \succeq 0$, then

$$A^{1/2} = Q \operatorname{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}) Q^\top$$

is symmetric PSD, $(A^{1/2})^2 = A$, and $A^{1/2}A = AA^{1/2}$. If $A \succ 0$, it and its square root are invertible, and $A^{-1} \succ 0$.

Proof. The expression $x^\top Ax = \sum_j \lambda_j c_j^2$ proves sufficiency. Testing $x = q_j$ proves necessity. Multiplying the proposed square root by itself cancels $Q^\top Q$ and gives A ; diagonal matrices commute, proving the commuting identity. For positive eigenvalues, their square roots and reciprocals are positive, so both inverse assertions follow from the same spectral test. $\square$

The uniqueness of the PSD square root is proved in the exercises. In particular, positivity is stronger than a positive determinant, which only specifies the product of eigenvalues.

Proposition 9.7 (Gram matrices and kernels). For every real $m \times n$ matrix B ,

$$B^\top B \succeq 0, \quad \ker(B^\top B) = \ker B, \quad \operatorname{rank}(B^\top B) = \operatorname{rank} B.$$

It is PD exactly when B has independent columns. Also $BB^\top \succeq 0$, with $\ker(BB^\top) = \ker B^\top$.

Proof. $x^\top B^\top Bx = \|Bx\|^2$. If $B^\top Bx = 0$, this squared norm is zero, so $Bx = 0$; the reverse implication is immediate. Rank-nullity, on the same n -dimensional domain, gives equal ranks. The squared norm is positive for all $x \neq 0$ precisely when $\ker B = \{0\}$. Replace B by $B^\top$ for the last assertion. $\square$

Example 9.4 (Worked reading: a rectangular Gram factor and a square root). For $D = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$,

$$D^\top D = L, \quad DD^\top = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

The equation $Dx = 0$ gives $x_1 = x_2 = x_3$, so L is PSD of rank two, while $DD^\top$, with eigenvalues 1, 3, is PD. The PSD square root of L is the square matrix

$$L^{1/2} = \frac{1}{2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix} + \frac{1}{2\sqrt{3}} \begin{pmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{pmatrix}.$$

The two summands are $q_1 q_1^\top$ and $\sqrt{3} q_3 q_3^\top$. Their cross-products vanish and each $q_j q_j^\top$ is idempotent, so squaring gives L . The rectangular factor $D^\top D = L$ and the square-root identity $(L^{1/2})^2 = L$ describe different factorizations.

5. Spectral projectors, rank, and trace

Proposition 9.8 (Projectors onto eigenspaces). For each distinct eigenvalue λ , let $P_\lambda = \sum_{j: \lambda_j = \lambda} q_j q_j^\top$. Then P_λ is the orthogonal projector onto E_λ , and

$$P_\lambda^\top = P_\lambda, \quad P_\lambda^2 = P_\lambda, \quad P_\lambda P_\mu = 0 \ (\lambda \neq \mu), \quad \sum_\lambda P_\lambda = I, \quad A = \sum_\lambda \lambda P_\lambda.$$

Furthermore

$$\operatorname{rank} A = \#\{j : \lambda_j \neq 0\}, \quad \operatorname{tr} A = \sum_j \lambda_j, \quad \operatorname{tr} P_\lambda = \operatorname{rank} P_\lambda = \dim E_\lambda.$$

Proof. $P_\lambda x$ keeps exactly the coordinates of x in the orthonormal basis of E_λ , proving the projection assertion. Multiplying sums and using $q_i^\top q_j = \delta_{ij}$ proves the identities. Similarity to the diagonal matrix preserves rank and gives the rank count. Finally $\text{tr}(q_j q_j^\top) = q_j^\top q_j = 1$; linearity of trace proves both trace formulas. $\square$

Example 9.5 (Worked reading: projector arithmetic). For L , its kernel projector and image projector are

$$P_0 = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad P_{\text{im } L} = I - P_0.$$

The first replaces the coordinates by their common average; the second subtracts that average. Their traces are 1 and 2, agreeing with the dimensions of the constant-vector line and its orthogonal plane. For $x = (1, 0, 0)^\top$, these components are $(1, 1, 1)^\top/3$ and $(2, -1, -1)^\top/3$. Since $Lq_0 = 0$, $Lq_1 = q_1$, and $Lq_3 = 3q_3$, its image is exactly the plane orthogonal to q_0 .

6. Symmetric differential equations

For $x'(t) = Ax(t)$, $A = Q\Lambda Q^\top$, set $y = Q^\top x$. Then $y' = \Lambda y$, so

$$x(t) = \sum_j e^{\lambda_j t} q_j q_j^\top x(0).$$

Every solution has this form: multiplying by $Q^\top$ reduces it to the scalar equations, whose solutions follow by differentiating $e^{-\lambda_j t} y_j(t)$.

For $x' = -Lx$, $x(0) = (1, 0, 0)^\top$, the three spectral coordinates at zero are $1/\sqrt{3}, 1/\sqrt{2}, 1/\sqrt{6}$, respectively. Hence

$$x(t) = \frac{1}{3}(1, 1, 1)^\top + \frac{e^{-t}}{2}(1, 0, -1)^\top + \frac{e^{-3t}}{6}(1, -2, 1)^\top.$$

At zero the sum is $(1, 0, 0)^\top$; differentiating multiplies the last two terms by $-1, -3$, exactly the action of $-L$. The first term is constant and the others approach zero as $t \rightarrow \infty$. Thus the limiting vector has all entries equal to the initial average.

Exercises and solutions

All matrices in the positive-form exercises are real and symmetric when a sign or order is asserted. A star marks an advanced source exercise. All vector spaces and matrices throughout the unit are real.

Real symmetric spectra

Exercise 9.1 (Symmetric orthogonal matrices). Prove that a real symmetric orthogonal matrix has eigenvalues only 1 and -1 . If it is also positive definite, prove it is I .

Solution. The real spectral theorem gives an eigenbasis. Since $A^\top A = I$ and $A^\top = A$, we have $A^2 = I$. For an eigenvector $v \neq 0$, $v = A^2 v = \lambda^2 v$, so $\lambda^2 = 1$. If A is PD, every eigenvalue is positive, hence 1. The real spectral theorem then gives $A = QIQ^\top = I$.

Exercise 9.2 (Independence and orthogonality with parameters). Let $A = \begin{pmatrix} a & 1 \\ ab & b \end{pmatrix}$, where real $a, b \neq 0$.

- Find the eigenvalues and normalized eigenvectors.
- Show that eigenvectors for the two roots are independent unless $a + b = 0$.
- Show that these directions are orthogonal exactly when $ab = 1$.

Solution. The polynomial is $t^2 - (a + b)t = t(t - a - b)$. Vectors $v_0 = (1, -a)^\top$ and $v_{a+b} = (1, b)^\top$ satisfy the eigenvector equations. Their unit versions are $v_0/\sqrt{1+a^2}$ and $v_{a+b}/\sqrt{1+b^2}$, with either sign. Their column determinant is $a + b$; if $a + b \neq 0$, they are independent. If $a + b = 0$, they coincide and the single eigenspace for zero is $\text{span}(1, -a)^\top$: $A \neq 0$ has rank one. Their dot product before normalization is $1 - ab$, proving (c). The condition $ab = 1$ is exactly the symmetry condition for A , consistent with the symmetric eigenvector theorem.

Exercise 9.3 (A repeated eigenvalue). Let λ have algebraic multiplicity $k \geq 2$ in a real symmetric matrix.

- Prove its eigenspace has an orthonormal basis of k vectors.
- Prove there are infinitely many such orthonormal bases.
- Prove there cannot be more than k independent eigenvectors for λ , and conclude the eigenspace has dimension k .

Solution. The spectral theorem diagonalizes the matrix. The k columns belonging to λ form an orthonormal basis of that eigenspace. If the first two are q_1, q_2 , replace them by $q_1 \cos t + q_2 \sin t$ and $-q_1 \sin t + q_2 \cos t$. They remain orthonormal and in the same eigenspace, giving infinitely many bases as t varies. The earlier geometric-multiplicity bound gives dimension at most k , while the k constructed vectors give dimension at least k .

Exercise 9.4 (An orthonormal matrix spanning a kernel). Let $A \in \mathbb{R}^{m \times n}$ have rank r .

- Use the eigenspaces of $A^\top A$ to construct $S \in \mathbb{R}^{n \times (n-r)}$ with $AS = 0$ and $S^\top S = I_{n-r}$.
- If $AT = 0$, prove $\text{rank } T \leq n - r$.

Solution. The Gram kernel identity gives $\ker(A^\top A) = \ker A$, of dimension $n - r$. Choose orthonormal eigenvectors for its zero eigenvalue and place them in S . They give both stated identities. Every column of T lies in $\ker A$, so its column-space dimension is at most $n - r$. When $r = n$, S has no columns and the identities use the empty zero-dimensional identity.

Exercise 9.5 (*Simultaneous orthogonal diagonalization). For real symmetric A, B of the same order, prove that one orthogonal matrix diagonalizes both if and only if $AB = BA$.

Solution. If $A = Q\Lambda Q^\top$, $B = QMQ^\top$ with diagonal Λ, M , their products commute because $\Lambda M = M\Lambda$. Conversely, commutation implies $B(E_\lambda(A)) \subseteq E_\lambda(A)$. On each such eigenspace, the restriction of B is self-adjoint. The spectral theorem gives an orthonormal basis there of B -eigenvectors, which are also A -eigenvectors. Different A -eigenspaces are orthogonal, so the union of these bases is an orthonormal basis of the whole space. Its column matrix diagonalizes both.

Exercise 9.6 (*Craig–Sakamoto identity). For real symmetric A, B of the same order, prove

$$AB = 0 \iff \det(I - sA) \det(I - tB) = \det(I - sA - tB) \quad \text{for every real } s, t.$$

Solution. If $AB = 0$, then $(I - sA)(I - tB) = I - sA - tB$; take determinants.

For the converse, work for all s in a sufficiently small interval about zero, where $C(s) = (I - sA)^{-1}$ exists. The assumed identity, divided by $\det(I - sA)$, gives $\det(I - tB) = \det(I - tC(s)B)$ for every t . Consequently B and $C(s)B$ have the same characteristic polynomial, and

$$h(s) := \operatorname{tr}\{C(s)BC(s)B\} = \operatorname{tr} B^2.$$

The trace-of-square conclusion follows directly from the coefficients of the determinant, without any hypothesis on its roots: the first coefficient gives $\operatorname{tr} M$, and the second gives $\{(\operatorname{tr} M)^2 - \operatorname{tr}(M^2)\}/2$, by summing the two-by-two principal minors. For order one, the first coefficient already gives M itself.

The entries of $C(s)$ are differentiable rational functions near zero. Differentiating $(I - sA)C(s) = I$ gives $C'(0) = A$, $C''(0) = 2A^2$. Twice applying the product rule to h , and cyclically moving factors inside the trace, yields

$$0 = h''(0) = 4 \operatorname{tr}(A^2 B^2) + 2 \operatorname{tr}(ABAB).$$

Put $M = AB$. Symmetry gives $\operatorname{tr}(A^2 B^2) = \operatorname{tr}(MM^\top) = \|M\|_F^2$, where $\|M\|_F^2 = \sum_{i,j} m_{ij}^2$. Also

$$\operatorname{tr}(M^2) + \|M\|_F^2 = \frac{1}{2} \|M + M^\top\|_F^2.$$

Therefore

$$0 = \operatorname{tr}(M^2) + 2 \|M\|_F^2 = \frac{1}{2} \|M + M^\top\|_F^2 + \|M\|_F^2.$$

Both terms are nonnegative, so $M = 0$. The proof does not assume $\operatorname{tr}(M^2) \geq 0$ for a nonsymmetric matrix; that assertion would be false.

Exercise 9.7 (The norm of a matrix acting on a vector). For real $A \in \mathbb{R}^{m \times n}$, let μ be the largest eigenvalue of $A^\top A$. Prove $\|Ax\| \leq \sqrt{\mu} \|x\|$ for every x .

Solution. The Gram matrix is PSD, so $\mu \geq 0$. For $x \neq 0$, the Rayleigh bound gives $\|Ax\|^2 = x^\top A^\top A x \leq \mu x^\top x$. Taking nonnegative square roots proves the result. For $x = 0$, both sides vanish. Equality for $x \neq 0$ occurs precisely in the top eigenspace of $A^\top A$.

Positive forms: basic tests and constructions

Exercise 9.8 (Quadratic-form uniqueness). (a) For real square A , show $x^\top A x = x^\top (A + A^\top) x / 2$.

(b) Prove $x^\top A x = 0$ for all x exactly when A is skew-symmetric.

(c) Deduce that two symmetric matrices with the same quadratic form are equal.

Solution. Transposing a scalar gives $x^\top Ax = x^\top A^\top x$, proving (a). If $A^\top = -A$, that scalar equals its negative and is zero. Conversely, setting $x = e_i$ gives $a_{ii} = 0$; setting $x = e_i + e_j$ gives $a_{ij} + a_{ji} = 0$. Thus $A^\top = -A$. If A, B are symmetric and have the same form, $A - B$ is both symmetric and skew-symmetric, hence zero.

Exercise 9.9 (Represent and classify three forms). Write the following as symmetric matrix forms and classify their signs:

$$x^2 + y^2 - xy, \quad 4x^2 + 5y^2 + z^2 + 2xy + 2yz, \quad -x^2 + 2y^2 + xy.$$

Solution. Their matrices are

$$A = \begin{pmatrix} 1 & -1/2 \\ -1/2 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & 1 & 0 \\ 1 & 5 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} -1 & 1/2 \\ 1/2 & 2 \end{pmatrix}.$$

The first form is $(x - y/2)^2 + 3y^2/4$, positive for nonzero (x, y) . The second is

$$4(x + y/4)^2 + (z + y)^2 + \frac{15}{4}y^2,$$

also positive for every nonzero vector. The third takes values -1 at $(1, 0)$ and 2 at $(0, 1)$, so it is indefinite.

Exercise 9.10 (Length preservation). For a fixed real square matrix Q , prove that the following are equivalent: $Q^\top Q = I$; $\|Qx\| = \|x\|$ for every x ; $(Qx)^\top (Qy) = x^\top y$ for every x, y . Does equality of lengths for just one vector imply orthogonality?

Solution. $Q^\top Q = I$ gives the inner-product identity, which gives norm preservation on taking $y = x$. Conversely, norm preservation makes $x^\top (Q^\top Q - I)x = 0$ for all x . The matrix is symmetric, so the preceding quadratic-form uniqueness result gives $Q^\top Q - I = 0$. A single vector is insufficient: $Q = \text{diag}(1, 2)$ preserves the length of e_1 but is not orthogonal. The matrix must be fixed and the condition must hold for all vectors.

Exercise 9.11 (Signs and diagonal matrices). (a) Prove $A \succ 0$ iff $-A \prec 0$, and $A \succeq 0$ iff $-A \preceq 0$.

(b) Can a matrix be both positive and negative definite? Both positive and negative semidefinite?

(c) For diagonal $D = \text{diag}(d_i)$, characterize PD and PSD.

Solution. Multiplication by -1 reverses the sign of every quadratic value, proving (a). A positive-dimensional matrix cannot have both strictly positive and strictly negative quadratic values on a fixed nonzero vector. Both weak signs hold exactly when its quadratic form vanishes, which for symmetric A means $A = 0$. For a diagonal matrix $x^\top Dx = \sum_i d_i x_i^2$. If all $d_i > 0$ it is PD; if all $d_i \geq 0$ it is PSD. Testing $x = e_i$ proves the necessity of those respective conditions.

Exercise 9.12 (*Entrywise bounds). Let $A = (a_{ij}) \succ 0$.

(a) Prove $a_{ii} > 0$.

(b) For $i \neq j$, prove $a_{ij}^2 < a_{ii}a_{jj}$.

(c) Prove a largest absolute entry of A is a diagonal entry.

(d) State and prove the corresponding PSD conclusions.

Solution. Testing e_i gives (a). For $x = te_i + e_j$, positivity gives $a_{ii}t^2 + 2a_{ij}t + a_{jj} > 0$ for every real t . Its minimum at $t = -a_{ij}/a_{ii}$ is $a_{jj} - a_{ij}^2/a_{ii} > 0$, proving (b). Therefore $|a_{ij}| < \sqrt{a_{ii}a_{jj}} \leq \max_k a_{kk}$. For PSD matrices replace the strict signs by weak signs. If $a_{ii} = 0$, the expression $2a_{ij}t + a_{jj}$ must be nonnegative for every t , which forces $a_{ij} = 0$. If $a_{ii} > 0$, minimize as before to obtain $a_{ij}^2 \leq a_{ii}a_{jj}$. These prove the diagonal maximum statement also in the PSD case.

Exercise 9.13 (Sums, trace, and determinant). Let $A \succeq 0$.

- (a) Prove $\operatorname{tr} A \geq 0$, with equality exactly when $A = 0$.
- (b) Prove $\det A \geq 0$, with equality exactly when A is singular.
- (c) If $B \succeq 0$, prove $A + B \succeq 0$; if $B \succ 0$, prove $A + B \succ 0$.

Solution. The eigenvalues λ_i are nonnegative. Their sum is nonnegative and vanishes exactly when every one is zero; spectral decomposition then gives $A = 0$. Their product is nonnegative and vanishes exactly when an eigenvalue is zero, equivalently when A is singular. Finally $x^\top(A + B)x = x^\top Ax + x^\top Bx$. The terms are nonnegative, and the second is strictly positive for $x \neq 0$ if $B \succ 0$, proving (c).

Exercise 9.14 (When a Gram matrix is zero). For real $A \in \mathbb{R}^{m \times n}$, prove $A^\top A, AA^\top \succeq 0$ and

$$A = 0 \iff A^\top A = 0 \iff \operatorname{tr}(A^\top A) = 0.$$

Solution. Their quadratic forms are $\|Ax\|^2$ and $\|A^\top y\|^2$. Also $\operatorname{tr}(A^\top A) = \sum_{i,j} a_{ij}^2$. This sum vanishes precisely when all entries vanish. The forward implications to a zero Gram matrix and zero trace are immediate, and the sum-of-squares formula proves the reverse implication.

Exercise 9.15 (Positive determinant and trace are insufficient). Prove that a PD matrix has positive determinant and positive trace. Show that these two scalar conditions together do not imply positive definiteness, even for a real symmetric matrix.

Solution. All eigenvalues of a PD matrix are positive, so their product and sum are positive. Conversely $A = \operatorname{diag}(-1, -1, 3)$ has determinant $3 > 0$ and trace $1 > 0$, but $e_1^\top A e_1 = -1 < 0$; it is indefinite.

Exercise 9.16 (Inverses and powers preserve positivity). (a) Prove a PD matrix is invertible, and a PSD matrix is PD exactly when it is invertible.

- (b) For symmetric invertible A , prove $A \succ 0$ iff $A^{-1} \succ 0$.
- (c) If A is PD or PSD, prove the same respective property for A^k , $k = 1, 2, \dots$

Solution. In the spectral representation, invertibility means that every eigenvalue is nonzero. A nonnegative eigenvalue list is strictly positive exactly when none is zero, proving (a). The eigenvalues of A^{-1} are $1/\lambda_i$, which are positive exactly when all λ_i are positive; the inverse is symmetric. The powers have the same orthonormal eigenvectors and eigenvalues λ_i^k , which retain positivity or nonnegativity.

Exercise 9.17 (Complete a matrix square). Let $A, B \succ 0$, $C = (A^{-1} + B^{-1})^{-1}$, and $D = A + B$.

- (a) Prove

$$D^{-1} = A^{-1} - A^{-1}CA^{-1} = B^{-1} - B^{-1}CB^{-1} = A^{-1}CB^{-1} = B^{-1}CA^{-1}.$$

- (b) Find μ and an x -independent quantity $Q(a, b)$ such that

$$(x - a)^\top A^{-1}(x - a) + (x - b)^\top B^{-1}(x - b) = (x - \mu)^\top C^{-1}(x - \mu) + Q(a, b).$$

- (c) Prove $Q(a, b) = (a - b)^\top D^{-1}(a - b)$.

Solution. All inverses exist: sums of PD matrices are PD, as are their inverses. Since $A^{-1}DB^{-1} = B^{-1} + A^{-1} = C^{-1}$, inverting gives $C = BD^{-1}A$. Interchanging A, B also gives $C = AD^{-1}B$. Multiplying these identities gives $A^{-1}CB^{-1} = B^{-1}CA^{-1} = D^{-1}$. Furthermore $C(A^{-1} + B^{-1}) = I$, so

$$A^{-1} - A^{-1}CA^{-1} = A^{-1}(I - CA^{-1}) = A^{-1}CB^{-1} = D^{-1};$$

the other formula follows by interchange.

Set $h = A^{-1}a + B^{-1}b$ and $\mu = Ch$. Expansion of both sides of (b) gives the same quadratic term $x^\top C^{-1}x$ and linear term $-2x^\top h$, provided

$$Q(a, b) = a^\top A^{-1}a + b^\top B^{-1}b - h^\top Ch.$$

Using (a), the coefficients of the pure a and pure b terms are D^{-1} , while the two cross terms are $-a^\top D^{-1}b - b^\top D^{-1}a$. Thus $Q(a, b) = (a - b)^\top D^{-1}(a - b)$, as claimed.

Exercise 9.18 (*A shift and a gap in the spectrum). (a) For symmetric A , show that some real k makes $kI + A$ PD.

(b) If symmetric A has no eigenvalue in $[a, b]$, where $a \leq b$, prove $(A - aI)(A - bI) \succ 0$.

Solution. Choose $k > -\lambda_{\min}(A)$; the shifted eigenvalues $k + \lambda_i$ are all positive. For (b), the two factors commute and are simultaneously diagonal in an eigenbasis of A . Their product has eigenvalues $(\lambda_i - a)(\lambda_i - b)$. Since $\lambda_i < a$ or $\lambda_i > b$, both factors have the same strict sign. Every product is positive, proving the claim, also when $a = b$.

Exercise 9.19 (A sum of two rank-one matrices). Let $a, b \in \mathbb{R}^n$ be independent and $A = aa^\top + bb^\top$. Prove $A \succeq 0$ with rank two. Find its two positive eigenvalues through a two-by-two equation, and express the corresponding eigenvectors as combinations of a, b .

Solution. Let $X = (a \ b)$. Then $A = XX^\top \succeq 0$, and the Gram rank identity gives $\text{rank } A = \text{rank } X = 2$. Its image is $\text{span}(a, b)$. A nonzero eigenvalue has eigenvector $x = \lambda^{-1}Ax$ in that image, so write $x = \theta_1 a + \theta_2 b$. Independence makes $Ax = \lambda x$ equivalent to

$$\begin{pmatrix} \lambda - a^\top a & -a^\top b \\ -a^\top b & \lambda - b^\top b \end{pmatrix} \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = 0.$$

The Gram matrix $X^\top X$ is PD, so its two eigenvalues are positive; they are the roots of

$$\lambda^2 - (\|a\|^2 + \|b\|^2)\lambda + \|a\|^2 \|b\|^2 - (a^\top b)^2 = 0.$$

Every nonzero solution θ produces a nonzero eigenvector $X\theta$. All remaining $n - 2$ eigenvalues are zero.

Exercise 9.20 (A difference of two rank-one matrices). Let $a, b \in \mathbb{R}^n$ be independent and $A = aa^\top - bb^\top$. Prove its rank is two and its two nonzero eigenvalues have opposite signs. Derive the equation

$$\begin{pmatrix} \lambda - a^\top a & a^\top b \\ -a^\top b & \lambda + b^\top b \end{pmatrix} \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = 0$$

and show the corresponding eigenvectors are $x = \theta_1 a - \theta_2 b$.

Solution. The image lies in $\text{span}(a, b)$. In the basis $a, -b$ of this plane, the restriction has matrix

$$K = \begin{pmatrix} \|a\|^2 & -a^\top b \\ a^\top b & -\|b\|^2 \end{pmatrix}.$$

Its determinant is $(a^\top b)^2 - \|a\|^2 \|b\|^2 < 0$, by strict Schwarz for independent vectors. Thus its restriction is invertible, so the rank is two. Its characteristic polynomial has real coefficients and negative constant term; its discriminant is the squared trace minus four times that negative determinant, hence positive. The two real roots have negative product and therefore opposite signs. The displayed equation is $(\lambda I - K)\theta = 0$, and $x = \theta_1 a - \theta_2 b$ translates it back to the original space. Thus the form is indefinite.

Exercise 9.21 (Factors of a PSD matrix). Let $A \succeq 0$ have order n and rank r .

- (a) Construct $T \in \mathbb{R}^{n \times r}$ and positive diagonal $\Lambda \in \mathbb{R}^{r \times r}$ with $A = TT^\top$, $T^\top T = \Lambda$.
- (b) For each $r \leq m \leq n$, construct an $n \times m$ matrix B with $A = BB^\top$.

Solution. Take the orthonormal eigenvectors $q_1, \dots, q_r$ for the positive eigenvalues and put $T = (\sqrt{\lambda_1}q_1 \ \cdots \ \sqrt{\lambda_r}q_r)$. Then $TT^\top = \sum_{j=1}^r \lambda_j q_j q_j^\top = A$, and $T^\top T = \text{diag}(\lambda_1, \dots, \lambda_r)$. If A is PD, $r = n$ and T is square and invertible. For (b), append $m - r$ zero columns to T . For $A = 0$, use the empty matrix when $r = m = 0$, or a zero matrix of the requested positive width.

Exercise 9.22 (Cholesky factorization in the definite case). Prove that every real $A \succ 0$ can be written $A = LL^\top$, where L is lower triangular with positive diagonal entries.

Solution. Induct on the order. For $n = 1$, take $L = [\sqrt{a_{11}}]$. Partition $A = \begin{pmatrix} a & b^\top \\ b & D \end{pmatrix}$, where $a > 0$. For $y \neq 0$, test the nonzero vector $(-b^\top y/a, y^\top)^\top$; positivity gives $y^\top(D - bb^\top/a)y > 0$. Thus $S = D - bb^\top/a$ is PD. By induction $S = L_2 L_2^\top$, with L_2 lower triangular and positive diagonal. Set

$$L = \begin{pmatrix} \sqrt{a} & 0 \\ b/\sqrt{a} & L_2 \end{pmatrix}.$$

Multiplication gives $LL^\top = A$, and every diagonal entry is positive. This proves precisely the positive-definite case.

Exercise 9.23 (*The unique PSD square root). (a) Construct a PSD B with $B^2 = A$ when $A \succeq 0$.

- (b) Prove this PSD B is unique, although unrestricted square roots need not be unique.
- (c) If $A \succ 0$, prove $(A^{-1})^{1/2} = (A^{1/2})^{-1}$.

Solution. An orthonormal eigenbasis gives $B = Q \text{diag}(\sqrt{\lambda_i}) Q^\top$, proving existence and providing a calculation method. If another PSD C satisfies $C^2 = A$, then $CA = AC$. Therefore each eigenspace $E_\lambda(A)$ is invariant under C . On this space C is self-adjoint and PSD, so it has an orthonormal eigenbasis with eigenvalues $\mu \geq 0$. The identity $C^2 = A$ gives $\mu^2 = \lambda$, hence $\mu = \sqrt{\lambda}$. Thus C acts as $\sqrt{\lambda}I$ on every $E_\lambda(A)$, exactly as B does; the eigenspaces span the whole space, so $C = B$. For example I has the unrestricted square roots I and $-I$, showing the PSD restriction matters. When $A \succ 0$, both matrices in (c) have eigenvectors q_i and eigenvalues $1/\sqrt{\lambda_i}$, proving their equality.

Exercise 9.24 (Matrix quadratic forms). Let $A \succeq 0$ be $n \times n$ and $B \in \mathbb{R}^{n \times m}$.

- (a) Prove $B^\top AB \succeq 0$ and $\text{rank}(B^\top AB) = \text{rank}(AB)$.
- (b) Prove $B^\top AB = 0$ iff $AB = 0$.
- (c) If $A \succ 0$, prove $\text{rank}(B^\top AB) = \text{rank } B$; deduce the conditions for $B^\top AB \succ 0$ and for $B^\top AB = 0$.
- (d) Even if B has full column rank $m < n$, does $B^\top AB \succ 0$ imply $A \succ 0$?

Solution. Set $C = A^{1/2}B$. Then $B^\top AB = C^\top C \succeq 0$, with kernel $\ker C$. Spectral decomposition gives $\ker A^{1/2} = \ker A$. Therefore $Cx = 0$ iff $ABx = 0$; equal kernels on the m -dimensional domain give equal ranks, proving (a). The rank is zero exactly when $AB = 0$, giving (b). If $A \succ 0$, multiplication by $A^{1/2}$ preserves rank, so the Gram matrix has rank $\text{rank } B$. It is PD exactly when B has rank m , and zero exactly when $B = 0$. For (d), take $A = \text{diag}(1, 0)$ and $B = (1, 0)^\top$. Then $B^\top AB = [1] \succ 0$, but A is singular.

Exercise 9.25 (Congruence preserves the sign of a form). If A is real symmetric and B is invertible of the same order, prove

$$B^\top AB \succ 0 \iff A \succ 0, \quad B^\top AB \succeq 0 \iff A \succeq 0.$$

Solution. For every x , $x^\top B^\top ABx = (Bx)^\top A(Bx)$. An invertible B maps all vectors bijectively onto all vectors and nonzero vectors onto nonzero vectors. The required strict or weak inequality therefore holds on one side exactly when it holds on the other. The two matrices need not have the same eigenvalues; this is a statement about signs of quadratic values.

Exercise 9.26 (Eigenvalues between zero and one). Let $A \succeq 0$, $B \succ 0$. Prove every eigenvalue of $(A + B)^{-1}A$ lies in $[0, 1)$.

Solution. Set $C = A + B \succ 0$. The given matrix is similar, via $C^{1/2}$, to $H = C^{-1/2}AC^{-1/2}$, which is symmetric PSD. Moreover $I - H = C^{-1/2}BC^{-1/2} \succ 0$. For a unit eigenvector of H , these two sign statements give $\lambda \geq 0$ and $1 - \lambda > 0$. Similarity preserves the eigenvalues, proving the result even though $(A + B)^{-1}A$ need not itself be symmetric.

Exercise 9.27 (Principal submatrices). Prove every principal submatrix of a PSD matrix is PSD, and every principal submatrix of a PD matrix is PD.

Solution. For selected coordinates indexed by S , let A_S be the principal submatrix. Extend a vector y on those coordinates to x by putting zeros in all other coordinates. Then $y^\top A_S y = x^\top A x$. Nonnegativity follows for every y , and $y \neq 0$ gives $x \neq 0$, so strict positivity is also inherited.

Exercise 9.28 (*Positive leading principal minors). Prove a real symmetric A is PD if and only if all leading principal minors are positive.

Solution. If A is PD, every leading principal submatrix is PD and has positive determinant, proving necessity. For sufficiency induct on n . The case $n = 1$ is immediate. Write $A = \begin{pmatrix} a & b^\top \\ b & D \end{pmatrix}$, with $a > 0$. For each leading principal submatrix S_k of $S = D - bb^\top/a$, the Schur determinant formula gives $\det S_k = \det A_{k+1}/a > 0$. By induction $S \succ 0$. For $x = (t, y^\top)^\top$,

$$x^\top A x = a \left(t + \frac{b^\top y}{a} \right)^2 + y^\top S y.$$

This is strictly positive if $y \neq 0$, or if $y = 0, t \neq 0$. Hence $A \succ 0$.

Exercise 9.29 (*All principal minors for semidefiniteness). Prove a real symmetric A is PSD if and only if all its principal minors, not merely the leading ones, are nonnegative.

Solution. Necessity follows because principal submatrices of a PSD matrix are PSD and their eigenvalue products are nonnegative. Conversely expand $\det(tI + A)$ by multilinearity in the columns. Selecting columns from A at positions in S , and te_j elsewhere, contributes $t^{n-|S|} \det A_S$. The cofactor sign for selecting the same omitted rows and columns is positive. Therefore

$$\det(tI + A) = \sum_{S \subseteq \{1, \dots, n\}} t^{n-|S|} \det A_S, \quad \det A_\emptyset = 1.$$

If all principal minors are nonnegative, this expression is strictly positive for $t > 0$, because it includes t^n . A negative eigenvalue λ of A would make it zero at $t = -\lambda > 0$, a contradiction. All eigenvalues are nonnegative, so $A \succeq 0$.

Exercise 9.30 (Small minors and one zero determinant). (a) For real a, b, p , classify

$$A = \begin{pmatrix} 0 & 0 \\ 0 & p \end{pmatrix}, \quad B = \begin{pmatrix} 1 & a & b \\ a & a^2 & ab \\ b & ab & p \end{pmatrix}$$

as PD or PSD, and list their principal minors. Explain why their leading principal minors are nonnegative for every p .

- (b) If a symmetric $n \times n$ matrix has positive leading principal minors of orders $1, \dots, n-1$, but determinant zero, prove it is PSD.

Solution. Neither displayed matrix is PD: $Ae_1 = 0$, and $B(a, -1, 0)^\top = 0$. For A , the principal minors are $0, p$ and 0 , so it is PSD exactly when $p \geq 0$. For B , the order-one minors are $1, a^2, p$; the order-two minors, on indices $12, 13, 23$, are $0, p - b^2, a^2(p - b^2)$, and its determinant is zero. Its quadratic form is $(x + ay + bz)^2 + (p - b^2)z^2$, hence it is PSD exactly when $p \geq b^2$. The leading minors are $0, 0$ for A , and $1, 0, 0$ for B , regardless of p .

For (b), the case $n = 1$ is the zero scalar matrix. For $n > 1$, partition as $M = \begin{pmatrix} C & b \\ b^\top & d \end{pmatrix}$, where $C \succ 0$ by the leading-minor criterion. The determinant formula gives $0 = \det C(d - b^\top C^{-1}b)$, so $d = b^\top C^{-1}b$. Then

$$(x^\top, t)M(x^\top, t)^\top = (x + tC^{-1}b)^\top C(x + tC^{-1}b) \geq 0.$$

Exercise 9.31 (Adding a PSD matrix increases determinant). If $A \succ 0$, $B \succeq 0$, prove $\det(A + B) \geq \det A$, with equality exactly when $B = 0$.

Solution. Set $C = A^{-1/2}BA^{-1/2} \succeq 0$. Factoring gives $\det(A+B) = \det A \det(I+C) = \det A \prod_i (1+\mu_i)$, where $\mu_i \geq 0$. Each factor is at least one. Equality holds exactly when all $\mu_i = 0$, which by spectral decomposition means $C = 0$, equivalently $B = 0$.

Exercise 9.32 (Real eigenvalues of a nonsymmetric product). If A, B are real symmetric and at least one is PSD, prove that p_{AB} splits into real linear factors.

Solution. Suppose $A \succeq 0$. Set $X = A^{1/2}$ and $Y = A^{1/2}B$. Then $XY = AB$, while $YX = A^{1/2}BA^{1/2}$ is symmetric. The equal-size product identity gives identical characteristic polynomials for XY, YX , including their zero roots. The real spectral theorem factors the characteristic polynomial of YX into real linear factors. The same is therefore true for AB , even when A is singular. If $B \succeq 0$, apply the argument to BA , which has the same characteristic polynomial as AB .

Exercise 9.33 (Simultaneous congruence with one positive form). Let $A \succ 0$ and $B = B^\top$ have the same order. Prove there are invertible T and real diagonal Λ such that

$$A = TT^\top, \quad B = T\Lambda T^\top,$$

where Λ contains the eigenvalues of $A^{-1}B$.

Solution. The matrix $H = A^{-1/2}BA^{-1/2}$ is real symmetric. Choose orthogonal Q with $H = Q\Lambda Q^\top$, and put $T = A^{1/2}Q$. Then $TT^\top = A$ and $T\Lambda T^\top = B$. Also $A^{1/2}(A^{-1}B)A^{-1/2} = H$, so $A^{-1}B$ is similar to H and has the eigenvalues on the diagonal of Λ .

Exercise 9.34 (Subtract a rank-one matrix). For $A \succ 0$ and a real column a , prove $A - aa^\top \succ 0$ if and only if $a^\top A^{-1}a < 1$.

Solution. Congruence by $A^{-1/2}$ turns the matrix into $I - uu^\top$, where $u = A^{-1/2}a$. If $u = 0$, this is I and the condition is $0 < 1$. Otherwise it acts as $1 - \|u\|^2$ on $\text{span } u$ and as 1 on $u^\perp$. Thus it is PD exactly when $\|u\|^2 = a^\top A^{-1}a < 1$.

Exercise 9.35 (Diagonal blocks and their order). (a) Prove $\text{diag}(A, D)$ is PD (respectively PSD) exactly when both A, D are PD (respectively PSD).

(b) Prove

$$\begin{pmatrix} A & B \\ B^\top & D \end{pmatrix} \succ 0 \iff \begin{pmatrix} D & B^\top \\ B & A \end{pmatrix} \succ 0.$$

Solution. For (a), the quadratic form is $x^\top Ax + y^\top Dy$. Both positive terms give the stated sign; necessity follows by setting one of x, y to zero. For (b), interchange the coordinate groups x, y . This is an invertible permutation of all nonzero vectors, and the quadratic value is unchanged. Equivalently the matrices are congruent by an orthogonal block permutation.

Exercise 9.36 (Fischer's determinant inequality). If $Z = \begin{pmatrix} A & B \\ B^\top & D \end{pmatrix} \succeq 0$, prove $\det Z \leq \det A \det D$.

Solution. First suppose $Z \succ 0$. Its principal block A is PD, and testing vectors $(-A^{-1}By, y^\top)^\top$ shows $S = D - B^\top A^{-1}B \succ 0$. The determinant formula gives $\det Z = \det A \det S$. Since $D = S + B^\top A^{-1}B$ and the second term is PSD, the determinant monotonicity result gives $\det D \geq \det S$. Multiplying by $\det A > 0$ proves the inequality.

For general PSD Z , apply this to $Z + \epsilon I$, $\epsilon > 0$. Its diagonal blocks are $A + \epsilon I, D + \epsilon I$, giving

$$\det(Z + \epsilon I) \leq \det(A + \epsilon I) \det(D + \epsilon I).$$

Each determinant is a polynomial in ϵ and hence continuous. Let ϵ decrease to zero to obtain the stated inequality, including singular diagonal blocks.

Exercise 9.37 (Positive Schur complements). For symmetric diagonal blocks in $Z = \begin{pmatrix} A & B \\ B^\top & D \end{pmatrix}$, prove that $Z \succ 0$ exactly when $A \succ 0$ and $D - B^\top A^{-1}B \succ 0$. Prove the equivalent criterion using D and $A - BD^{-1}B^\top$.

Solution. When $A \succ 0$, complete the square:

$$(x^\top, y^\top)Z(x^\top, y^\top)^\top = (x + A^{-1}By)^\top A(x + A^{-1}By) + y^\top(D - B^\top A^{-1}B)y.$$

If both named blocks are PD, the right side is positive for every nonzero pair (x, y) : either $y \neq 0$, or $y = 0, x \neq 0$. Conversely $Z \succ 0$ implies $A \succ 0$ by principal submatrices, and choosing $x = -A^{-1}By$, $y \neq 0$, makes the second form strictly positive. Interchanging the two coordinate groups gives the criterion using D .

Exercise 9.38 (*Positive block matrices and contractions). For real symmetric A of order m , D of order n , and $B \in \mathbb{R}^{m \times n}$, prove that

$$\begin{pmatrix} A & B \\ B^\top & D \end{pmatrix} \succeq 0$$

if and only if $A, D \succeq 0$ and there is $C \in \mathbb{R}^{m \times n}$ with $B = A^{1/2}CD^{1/2}$ and $C^\top C \preceq I_n$.

Solution. For sufficiency, put $u = A^{1/2}x$, $v = D^{1/2}y$. The block quadratic form is

$$\|u\|^2 + 2u^\top Cv + \|v\|^2 = \|u + Cv\|^2 + v^\top(I - C^\top C)v \geq 0.$$

For necessity, the principal blocks are PSD. If $x \in \ker A$, testing (tx, y) for all real t gives $2tx^\top By + y^\top Dy \geq 0$, hence $B^\top x = 0$. Likewise $y \in \ker D$ implies $By = 0$. Take orthonormal matrices U, V spanning the positive eigenspaces of A, D , and write $A = U\Delta U^\top$, $D = V\Gamma V^\top$, where the diagonal entries of Δ, Γ are positive. The kernel relations give $B = UU^\top BVV^\top$. If either positive eigenspace is zero, then $B = 0$, and $C = 0$ works.

Otherwise define $C_0 = \Delta^{-1/2}U^\top B V \Gamma^{-1/2}$. Restrict the original quadratic form to these eigenspaces and scale coordinates by the inverse square roots. This gives $\begin{pmatrix} I & C_0 \\ C_0^\top & I \end{pmatrix} \succeq 0$. Testing $(-C_0 z, z)$ yields $z^\top(I - C_0^\top C_0)z \geq 0$. Set $C = UC_0V^\top$. Then $A^{1/2}CD^{1/2} = B$, and

$$\|Cy\|^2 = \|C_0V^\top y\|^2 \leq \|V^\top y\|^2 \leq \|y\|^2.$$

Thus $C^\top C \preceq I_n$. No inverse of a singular diagonal block has been used.

Exercise 9.39 (*A bordered Gram matrix). Let $A \succeq 0$ have order m , $B \in \mathbb{R}^{m \times n}$, and

$$Z = \begin{pmatrix} A & B \\ B^\top & 0 \end{pmatrix}, \quad G = A + BB^\top.$$

Give an example where A is singular and Z is invertible, and state whether G is singular in that example. Prove that Z is invertible exactly when B has full column rank and $G \succ 0$.

Solution. Take scalar blocks $A = 0, B = 1$. Then Z exchanges the two coordinates and is invertible, while $G = 1$ is also invertible.

In general $G \succeq 0$, and $u^\top Gu = u^\top Au + \|B^\top u\|^2$. For a PSD matrix, $u^\top Au = 0$ iff $Au = 0$, by its spectral representation. Therefore $\ker G = \ker A \cap \ker B^\top$. The equations $Z(u, v)^\top = 0$ are $Au + Bv = 0$, $B^\top u = 0$. Multiplying the first by $u^\top$ gives $u^\top Au = 0$, hence $Au = 0$ and $Bv = 0$. Thus

$$\ker Z = (\ker G) \times (\ker B).$$

It is zero precisely when G is invertible, equivalently PD, and B has full column rank. The same kernel decomposition gives the stronger rank formula $\text{rank } Z = \text{rank } G + \text{rank } B$.

Exercise 9.40 (Inverse and determinant of the bordered matrix). Use A, B, Z, G as in the preceding problem.

(a) If Z is invertible, put $H = B^\top G^{-1}B$. Prove $H \succ 0$ and

$$Z^{-1} = \begin{pmatrix} G^{-1} - G^{-1}BH^{-1}B^\top G^{-1} & G^{-1}BH^{-1} \\ H^{-1}B^\top G^{-1} & I - H^{-1} \end{pmatrix}.$$

(b) Assuming only G is invertible, prove $\det Z = (-1)^n \det G \det(B^\top G^{-1}B)$.

(c) If A is invertible, prove $\det Z = (-1)^n \det A \det(B^\top A^{-1}B)$. Deduce, for PD A , the equality of the two products without the factor $(-1)^n$.

Solution. For (a), invertibility of Z gives $G \succ 0$ and full column rank of B , so $H \succ 0$ by the matrix quadratic-form result. Write the proposed blocks as $M, N, N^\top, K$, in order. Then $B^\top M = 0$, $B^\top N = I$. Using $A = G - BB^\top$,

$$AM = GM = I - BN^\top, \quad AN + BK = BH^{-1} - B + B - BH^{-1} = 0.$$

Thus multiplication by Z on the left gives the identity. Both matrices are symmetric, so transposing gives the other product as well. This proves the inverse formula.

For (b), add B times the second block row of Z to its first. This leaves the determinant unchanged and gives $\begin{pmatrix} G & B \\ B^\top & 0 \end{pmatrix}$. Its Schur complement is $-B^\top G^{-1}B$, proving (b). No invertibility of that Schur complement is required. If A is invertible, apply the same Schur formula directly to Z , giving (c). For $A \succ 0$, $G = A + BB^\top \succ 0$, so both formulas apply to the same determinant. Cancelling $(-1)^n$ gives

$$\det G \det(B^\top G^{-1}B) = \det A \det(B^\top A^{-1}B),$$

even if B is not full column rank and both sides vanish.

Idempotents and projections

In this section $P^2 = P$ means that P is idempotent. Matrices are real; symmetry is imposed only when stated.

- Exercise 9.41** (Basic idempotent operations). (a) Determine the diagonal entries of a diagonal idempotent.
- (b) If P is idempotent, show $P^\top, P^k$ for $k \geq 1$, and $I - P$ are idempotent. When is $-P$ idempotent?
- (c) Show $\text{diag}(P, Q)$ is idempotent exactly when both diagonal blocks are idempotent.

Solution. A diagonal entry satisfies $d^2 = d$, so it is 0 or 1. Transposing $P^2 = P$ gives $(P^\top)^2 = P^\top$. Induction gives $P^k = P$, while $(I - P)^2 = I - 2P + P^2 = I - P$. For $-P$, the required identity is $P = -P$, equivalent over the reals to $P = 0$. Finally block multiplication gives $\text{diag}(P, Q)^2 = \text{diag}(P^2, Q^2)$, proving the last equivalence.

Exercise 9.42 (Nonsymmetric examples and a false converse).

- (a) Construct a nonsymmetric idempotent of rank greater than one.
- (b) Prove an idempotent's eigenvalues belong to $\{0, 1\}$. Does an eigenvalue list contained in $\{0, 1\}$ imply idempotence?

Solution. Take $P = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. Its square is itself, its first two columns are independent and its image is their span, so its rank is two; it is not symmetric. If $Pv = \lambda v$, then $Pv = P^2v = \lambda^2v$, giving $\lambda^2 = \lambda$. Conversely $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has only eigenvalue 1, but its square is $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, so it is not idempotent. The missing property is diagonalizability, which idempotence itself does imply.

Exercise 9.43 (Ordering orthogonal projectors). If P, Q are symmetric idempotents of the same order, prove $P \succeq Q$ if and only if $PQ = Q$.

Solution. Suppose $P - Q \succeq 0$. If $u \in \text{im } Q$, then $Qu = u$, so $0 \leq u^\top(P - Q)u = \|Pu\|^2 - \|u\|^2 \leq 0$. The last inequality is the Pythagorean decomposition $u = Pu + (I - P)u$ for an orthogonal projector. Equality forces $(I - P)u = 0$; hence $Pu = u$ on $\text{im } Q$, giving $PQ = Q$. Conversely, $PQ = Q$ and transposition give $QP = Q$. Thus $(P - Q)^2 = P - Q$, and $P - Q$ is symmetric idempotent. Its eigenvalues are 0, 1, so it is PSD.

Exercise 9.44 (Diagonal form and trace of an idempotent). Let P be an idempotent of order n .

- (a) Prove $\mathbb{R}^n = \text{im } P \oplus \ker P$, and construct a similarity $T^{-1}PT = \text{diag}(I_r, 0)$.
- (b) Construct an orthogonal, matrix S with $S^\top PS = \begin{pmatrix} I_r & C \\ 0 & 0 \end{pmatrix}$.
- (c) Prove $\text{rank } P = \text{tr } P = r$, where r is the algebraic multiplicity of eigenvalue 1.
- (d) Prove $\text{rank } n$ is equivalent to $P = I$, and all eigenvalues zero is equivalent to $P = 0$.

Solution. Every vector decomposes as $x = Px + (x - Px)$, with first term in the image and second in the kernel. If $y = Pz$ and $Pz = 0$, then $y = P^2z = Py = 0$, proving directness. Take T 's first columns as a basis of the image, on which P acts as the identity, and the remaining columns as a basis of the kernel, on which it acts as zero. This proves (a), with $r = \text{rank } P$.

For (b), choose instead an orthonormal basis of the image and extend it to an orthonormal basis of the whole space. The first columns are fixed by P , and every remaining column is mapped into the image. Therefore its matrix has exactly the displayed form. The diagonalization in (a) gives characteristic polynomial $(t - 1)^r t^{n-r}$ and trace r , since trace is similarity invariant. This proves (c). If rank is n , the image is the whole space and P acts as the identity. If all eigenvalues are zero, $r = 0$, so the rank is zero and $P = 0$. Both converses are immediate.

Exercise 9.45 (A rank characterization of idempotence). For any real square A of order n , prove

$$A^2 = A \iff \text{rank } A + \text{rank}(I - A) = n.$$

In particular this holds for symmetric A .

Solution. If $A^2 = A$, then $\text{im}(I - A) = \ker A$, by the decomposition in the preceding problem. Rank-nullity gives the rank equality. Conversely $x = Ax + (I - A)x$ shows $\text{im } A + \text{im}(I - A) = \mathbb{R}^n$. If the dimensions sum to n , the intersection is zero. But for every x , $A(I - A)x = (I - A)Ax$ belongs to both images, so it is zero. Hence $A - A^2 = 0$.

Exercise 9.46 (When a sum is idempotent). Let P, Q be idempotents of the same order.

(a) Prove $P + Q$ is idempotent exactly when $PQ = QP = 0$.

(b) In that case, prove $\text{rank}(P + Q) = \text{rank } P + \text{rank } Q$.

Solution. Expanding gives $(P + Q)^2 = P + Q + PQ + QP$. If the sum is idempotent, $PQ + QP = 0$. Multiplying this equation by P on the left and on the right gives $PQ + PQP = 0$ and $PQP + QP = 0$, respectively. Subtracting yields $PQ = QP$; combined with their zero sum, this gives $PQ = QP = 0$. The converse follows from the expansion. All three matrices are now idempotent, so rank equals trace for each:

$$\text{rank}(P + Q) = \text{tr}(P + Q) = \text{tr } P + \text{tr } Q = \text{rank } P + \text{rank } Q.$$

No symmetry was used.

Exercise 9.47 (Two absorption identities). If $AB = A$ and $BA = B$, prove that both A, B are idempotent.

Solution. Associativity gives $A^2 = (AB)A = A(BA) = AB = A$. Likewise $B^2 = (BA)B = B(AB) = BA = B$.

Exercise 9.48 (Two complementary column spaces). Let $A = (A_1 \ A_2)$ be square of order n , with $\text{rank } A_1 + \text{rank } A_2 = n$ and $A_1^\top A_2 = 0$. Prove

$$A_1(A_1^\top A_1)^{-1}A_1^\top + A_2(A_2^\top A_2)^{-1}A_2^\top = I_n.$$

Solution. If the respective column counts are $k, n - k$, their ranks are at most these counts; the rank sum forces equality in each. Thus both Gram matrices are invertible. The column spaces are orthogonal by $A_1^\top A_2 = 0$, and their dimensions sum to n ; hence their orthogonal direct sum is $\mathbb{R}^n$. Each displayed term is the orthogonal projector onto its corresponding column space. Their sum therefore fixes every vector and equals I_n . The projector formula follows directly by solving the normal equations for projection, or by an orthonormal basis of each column space. Empty column groups may be omitted from the formula.

Extrema and complete numerical examples

Exercise 9.49 (Convex and concave extremal eigenvalues). For symmetric A, B of order n and $0 \leq \alpha \leq 1$, prove

$$\lambda_{\max}(\alpha A + (1 - \alpha)B) \leq \alpha \lambda_{\max}(A) + (1 - \alpha) \lambda_{\max}(B),$$

$$\lambda_{\min}(\alpha A + (1 - \alpha)B) \geq \alpha \lambda_{\min}(A) + (1 - \alpha) \lambda_{\min}(B).$$

Solution. For every unit x , the Rayleigh bounds give

$$\alpha\lambda_{\min}(A) + (1 - \alpha)\lambda_{\min}(B) \leq x^\top \{\alpha A + (1 - \alpha)B\}x \leq \alpha\lambda_{\max}(A) + (1 - \alpha)\lambda_{\max}(B).$$

Taking the minimum in the first inequality and maximum in the second proves the two claims. The nonnegative weights are essential. Thus the largest eigenvalue is a convex function of the symmetric matrix, and the smallest is concave.

Exercise 9.50 (Extrema on a circle). Find the eigenvalues and maximum of the quadratic form on the unit circle for

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Solution. $p_A(t) = (t-2)^2 - 1 = (t-1)(t-3)$. Unit eigenvectors are $(1, 1)^\top/\sqrt{2}$ for 1 and $(1, -1)^\top/\sqrt{2}$ for 3. The maximum is 3, attained at both signs of the latter vector.

For B , $p_B(t) = t(t-1) - 1 = t^2 - t - 1$, giving $\lambda_\pm = (1 \pm \sqrt{5})/2$. A vector for λ is $(\lambda, 1)^\top$, because $(\lambda + 1, \lambda)^\top = (\lambda^2, \lambda)^\top$. The maximum is λ_+ , attained at $\pm(\lambda_+, 1)^\top/\sqrt{\lambda_+^2 + 1}$.

Exercise 9.51 (Extrema on a sphere). Find the eigenvalues and maximum of the associated quadratic

form on the unit sphere for $T = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$.

Solution. Expansion gives

$$p_T(t) = (t-2)\{(t-2)^2 - 1\} - (t-2) = (t-2)\{(t-2)^2 - 2\}.$$

The eigenvalues are $2 - \sqrt{2}, 2, 2 + \sqrt{2}$, with unit eigenvectors

$$\frac{1}{2}(1, \sqrt{2}, 1)^\top, \quad \frac{1}{\sqrt{2}}(1, 0, -1)^\top, \quad \frac{1}{2}(1, -\sqrt{2}, 1)^\top,$$

respectively. Direct multiplication verifies the equations; their pairwise dot products are zero. The maximum is $2 + \sqrt{2}$, attained at both signs of the last vector.

Exercise 9.52 (A form with positive and negative extrema). Find the maximum and minimum of $3x^2 + 5xy - 4y^2$ on $x^2 + y^2 = 1$.

Solution. The symmetric matrix is $A = \begin{pmatrix} 3 & 5/2 \\ 5/2 & -4 \end{pmatrix}$, with $p_A(t) = t^2 + t - 73/4$. Thus the extrema are the eigenvalues

$$\lambda_{\max} = \frac{-1 + \sqrt{74}}{2}, \quad \lambda_{\min} = \frac{-1 - \sqrt{74}}{2}.$$

Eigenvectors for these values are $v_\pm = (5, -7 \pm \sqrt{74})^\top$: the first equation $3x + (5/2)y = \lambda x$ holds directly, and the second follows from the characteristic equation. Normalize each $v_\pm$; the maximum and minimum are attained at both signs of the respective unit vectors. Their dot product is $25 + 49 - 74 = 0$, as symmetry predicts.

Exercise 9.53 (Symmetric idempotents and their factors). (a) Prove that a real symmetric matrix whose eigenvalues all belong to $\{0, 1\}$ is idempotent.

(b) If P is a symmetric idempotent of rank r , prove $P \succeq 0$ and construct an $n \times r$ matrix S with $S^\top S = I_r$, $P = SS^\top$.

Solution. The spectral theorem gives $P = Q\Lambda Q^\top$. For eigenvalues 0, 1, $\Lambda^2 = \Lambda$, so $P^2 = P$. Conversely an idempotent's eigenvalues satisfy $\lambda^2 = \lambda$, so a symmetric idempotent is PSD. Take S to consist of the orthonormal eigenvectors for eigenvalue 1. There are r such vectors, and $SS^\top = \sum_{\lambda_j=1} q_j q_j^\top = P$. For $r = 0$, this is the empty factorization of the zero matrix.

Functions of matrices: reading practice

For the following problems, a matrix power series is interpreted by convergence of every entry. The usual scalar series for the exponential, logarithm, and binomial function are assumed from calculus wherever they are invoked.

Exercise 9.54 (Functions of diagonal matrices). (a) If $D = \text{diag}(d_1, \dots, d_n)$ and the series converges, prove

$$\sum_{j=0}^{\infty} c_j D^j = \text{diag} \left(\sum_{j=0}^{\infty} c_j d_1^j, \dots, \sum_{j=0}^{\infty} c_j d_n^j \right).$$

Deduce $\exp(I_n) = eI_n$.

(b) For $A = \text{diag}(a, b)$, compute $\exp A$. If real $|a|, |b| < 1$, compute $(I + A)^\nu$ and $\log(I + A)$ from their scalar series.

(c) If instead $|a|, |b| > 1$ and ν is an integer, compute $(I + A)^\nu$.

Solution. Every finite partial sum is diagonal with the stated scalar partial sums on its diagonal. Taking their limits proves (a). Using the exponential series gives $\exp(I_n) = eI_n$, and $\exp A = \text{diag}(e^a, e^b)$. For $|a|, |b| < 1$, $1 + a, 1 + b > 0$ and the scalar series give

$$(I + A)^\nu = \text{diag}((1 + a)^\nu, (1 + b)^\nu), \quad \log(I + A) = \text{diag}(\log(1 + a), \log(1 + b)).$$

In (c), repeated diagonal multiplication gives the same power formula for nonnegative integers. Since $|a|, |b| > 1$ excludes $a = b = -1$ individually, both $1 + a, 1 + b$ are nonzero; inverse diagonal multiplication extends the formula to negative integers. No binomial-series convergence is claimed outside its scalar interval.

Exercise 9.55 (Nilpotence terminates a power series). If $A^k = 0$, prove that every formal series $\sum_{j=0}^{\infty} c_j A^j$ reduces to a polynomial of degree at most $k - 1$.

Solution. For $j \geq k$, $A^j = A^k A^{j-k} = 0$. Thus every partial sum from index $k - 1$ onward equals $c_0 I + c_1 A + \dots + c_{k-1} A^{k-1}$. The matrix series therefore requires no convergence assumption on the coefficients beyond these finitely many terms.

Exercise 9.56 (Functions of an idempotent). For an idempotent P , prove

$$(I + P)^{-1} = I - \frac{1}{2}P, \quad \exp P = I + (e - 1)P, \quad \log(I + P) = (\log 2)P.$$

Solution. In either multiplication order, $(I + P)(I - P/2) = I + P/2 - P^2/2 = I$. For every $j \geq 1$, $P^j = P$, so the exponential series gives

$$\exp P = I + \left(\sum_{j=1}^{\infty} \frac{1}{j!} \right) P = I + (e - 1)P.$$

The logarithm series has partial sums $\sum_{j=1}^N (-1)^{j+1} P^j / j = \{ \sum_{j=1}^N (-1)^{j+1} / j \} P$. The scalar alternating harmonic series converges to $\log 2$; hence every matrix entry converges to that of $(\log 2)P$. Symmetry is not needed.

Exercise 9.57 (A geometric matrix series). Let $I - A$ be invertible.

(a) For each integer $k \geq 0$, prove

$$(I - A)^{-1} = (I - A)^{-1} A^k + \sum_{j=0}^{k-1} A^j,$$

where the sum is empty for $k = 0$.

- (b) Assume in addition that A is symmetric and each of its real eigenvalues lies strictly between -1 and 1 . Prove $(I - A)^{-1} = \sum_{j=0}^{\infty} A^j$.

Solution. The telescoping identity $(I - A) \sum_{j=0}^{k-1} A^j = I - A^k$ gives (a) after multiplication by the inverse. For (b), write $A = Q\Lambda Q^\top$ using the real spectral theorem. Then $A^k = Q \operatorname{diag}(\lambda_1^k, \dots, \lambda_n^k) Q^\top$. Every λ_i^k tends to zero, since $|\lambda_i| < 1$. Each matrix entry is a fixed finite linear combination of these powers, so A^k tends entrywise to zero. Letting $k \rightarrow \infty$ in (a) proves the asserted convergence. The additional assumptions already imply invertibility of $I - A$, since none of its eigenvalues $1 - \lambda_i$ is zero.

Exercise 9.58 (Real orthogonal matrices as rotation blocks). Let $R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$.

- (a) For orthogonal S , prove $S \operatorname{diag}(R(\theta_1), \dots, R(\theta_k)) S^\top$ is orthogonal with determinant 1.
 (b) Prove the same after appending a final scalar block 1.
 (c) Prove these forms represent every real orthogonal matrix of determinant 1, in even and odd order respectively.

Solution. Each rotation block is orthogonal with determinant 1. Block multiplication and orthogonal similarity preserve these facts, including an appended scalar block 1. This proves (a,b).

For (c), let A be real orthogonal and put $H = (A + A^\top)/2$. The matrix H is real symmetric, so its mutually orthogonal eigenspaces span $\mathbb{R}^n$. Since $AH = HA = (A^2 + I)/2$, each eigenspace $E_c(H)$ is invariant under A . For a unit $u \in E_c(H)$,

$$c = u^\top H u = u^\top A u, \quad |c| \leq \|u\| \|Au\| = 1.$$

On $E_c(H)$, write $A = cI + K$. The identity $A + A^\top = 2cI$ shows $K^\top = -K$ on this space. Orthogonality gives

$$I = (cI - K)(cI + K) = c^2 I - K^2, \quad K^2 = -(1 - c^2)I.$$

If $c = \pm 1$, then $\|Ku\|^2 = -u^\top K^2 u = 0$, so $K = 0$ and $A = cI$. If $|c| < 1$, put $s = \sqrt{1 - c^2} > 0$. For any unit u in this space, set $v = Ku/s$. Then

$$\|v\| = 1, \quad u^\top v = 0, \quad Ku = sv, \quad Kv = -su.$$

Thus in the orthonormal basis (u, v) this invariant plane has matrix $\begin{pmatrix} c & -s \\ s & c \end{pmatrix} = R(\theta)$. Its orthogonal complement in $E_c(H)$ is K -invariant: for z perpendicular to the plane and w in it, $w^\top Kz = -(Kw)^\top z = 0$. Repeat on that complement. This constructs all rotation planes using only real vectors. The remaining directions carry $+1$ or -1 . There are an even number of -1 directions because $\det A = 1$; pair them into $R(\pi)$. Pair $+1$ directions into $R(0)$. Exactly one $+1$ direction remains when the total order is odd. Collecting these orthonormal bases proves (c).

Exercise 9.59 (Canonical blocks of a real skew-symmetric matrix). Let A be real and $A^\top = -A$.

- (a) Prove that an orthogonal S exists with

$$S^\top A S = \operatorname{diag} \left(\begin{pmatrix} 0 & -\theta_1 \\ \theta_1 & 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 & -\theta_k \\ \theta_k & 0 \end{pmatrix}, 0 \right),$$

for real $\theta_j > 0$ and $k \leq n/2$; the zero block may have size zero.

- (b) Deduce $\det A = 0$ in odd order and $\det A \geq 0$ in even order.

Solution. The matrix $-A^2 = A^\top A$ is real symmetric positive semidefinite. If it is zero, $\|Ax\|^2 = x^\top (-A^2)x = 0$ for every x , so $A = 0$. Otherwise the real spectral theorem supplies a unit u with $-A^2u = \theta^2u$, $\theta > 0$. Set $v = Au/\theta$. Then

$$\|v\|^2 = u^\top (-A^2)u/\theta^2 = 1, \quad u^\top v = u^\top Au/\theta = 0, \quad Au = \theta v, \quad Av = -\theta u.$$

The plane spanned by u, v is invariant and has the required block. Its orthogonal complement is invariant too: $w^\top Az = -(Aw)^\top z = 0$ for w in the plane and z perpendicular to it. Repeat for the skew-symmetric restriction to that smaller space. If its squared Gram matrix becomes zero, the restriction itself is zero by the first argument. Finite induction yields (a). Every nonzero two-by-two block has determinant $\theta_j^2 > 0$. A positive-size zero block makes the determinant zero. Odd order necessarily leaves such a block, whereas even order gives either zero or a product of positive numbers. This proves (b).

Exercise 9.60 (A power series in a symmetric matrix). Let $A = Q\Lambda Q^\top$ be real symmetric with Q orthogonal. Suppose the scalar series $f(t) = \sum_{j=0}^{\infty} c_j t^j$ converges at every eigenvalue of A . Prove that the matrix series converges and

$$\sum_{j=0}^{\infty} c_j A^j = Q \operatorname{diag}(f(\lambda_1), \dots, f(\lambda_n)) Q^\top.$$

Solution. Cancellation of $Q^\top Q$ gives $A^j = Q\Lambda^j Q^\top$ for every $j \geq 0$, including $j = 0$. Each finite partial sum therefore equals Q times the diagonal matrix of the scalar partial sums, times $Q^\top$. There are only finitely many diagonal positions, each converging by assumption. Each entry of the product is a fixed finite linear combination of them, so it converges to the stated value. Within a repeated eigenspace the value is the same scalar $f(\lambda)$; thus changing its orthonormal eigenbasis does not change the resulting matrix.